

A Tool based Experiment to Teach Elicitation and Specification of Virtual Reality Product Requirements

Sai Anirudh Karre
SERC, IIIT Hyderabad, India
saianirudh.karre@research.iiit.ac.in

Karthik Vaidhyanathan
SERC, IIIT Hyderabad, India
karthik.vaidhyanathan@iiit.ac.in

Y. Raghu Reddy
SERC, IIIT Hyderabad, India
raghu.reddy@iiit.ac.in

ABSTRACT

Students need to understand the assessment of requirements correctness while building software systems. It helps produce products that meet the stakeholder objectives. This poster illustrates an experiment conducted as a tool-based collaborative assignment for assessing the correctness of software products using a Virtual Reality (VR) application as part of a CS300-level course at our University.

CCS CONCEPTS

• **Applied computing** → **Collaborative learning**; • **Software and its engineering** → **Specification languages**;

KEYWORDS

Requirements Specification, Requirement Elicitation, Virtual Reality, Software Teams, Collaborative Assignment

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1 ABOUT THE COURSE

At IIIT Hyderabad, we offered a CS300 level Software Engineering (SE) course during Spring 2023 focused on building software systems using SE practices. The course had 130+ Undergraduate, Graduate, and Industry student learners. We randomly created 28 teams with a team size of 4 to 5 students for all the course related tasks and activities. Requirement engineering concepts were announced as part of the course to the students before announcing the assignment.

2 THE ASSIGNMENT

We conducted a team-based assignment as an experiment in a classroom setup to use the VReqST [3] tool to elicit and specify requirements using a VR application as an example. We chose a VR as it is still an emerging domain, and most of the students had not worked on VR before. As a prerequisite, we let students use some VR applications at our University VR lab and recommended

watching a few VR movies. The experiment consists of two phases, as described below. The duration of each phase is 2 weeks.

- **Phase 1:** Each team must internally elicit one VR application and write detailed requirements using a Software Requirement Specification (SRS) template [2] supplied to them.
- **Phase 2:** The SRS documents from the previous phase are exchanged randomly among teams. Each team is now required to use the VReqST tool [3] to elicit and specify the requirements of VR applications from the allocated SRS document, i.e., each team receives one random SRS document except the one they authored.

We developed VReqST, a requirement specification tool [3], based on the role-based model template for specifying VR software [1]. It allows specifying bare-minimum attributes of VR applications like Scenes, Articles, Actions-Reponses, Behaviors of Articles, and Timeline of overall Scenes with higher precision in JSON format. We created a well-defined evaluation rubric [2] to grade the requirements correctness of VR application by comparing SRS submission and its related VReqST specification. We have set up a face-time between the teams who authored the SRS submission and its correlated VReqST submission to grade the requirement correctness in the presence of a VR domain expert.

3 OBSERVATIONS

We observed 92% of VReqST submissions are valid and well-grounded compared to their correlated SRS specifications. 98% of VReqST submissions furnished accurate and correct placeholders to describe scene and article properties that are inadequately mentioned in SRS specifications. 91% of article behavior(s) and scene control flow are reasonably precise and explicit in VReqST Submissions. We also approached all the students to seek their experiences with this experiment. 87% of students revealed high-confidence in eliciting and specifying VR application requirements. 93% of students expressed that cross-evaluation of SRS submission with correlated VReqST submission helped them groom requirements. 96% of the students found the VReqST tool helped them understand VR applications in less time. This experiment is easy to replicate, and we encourage computing instructors to conduct this experiment using our resources [3] to teach requirement gathering of VR applications.

REFERENCES

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